Workshop equipment

SENSE + THINK

Arduino Nicla Vision

Days 1–3 · vision and audio models



- **STM32H747** dual core: Cortex-M7 480 MHz + M4 240 MHz
- 2 MP camera, MEMS microphone, 6-axis IMU, distance sensor
- Wi-Fi and Bluetooth LE

TRANSMIT

Grove Wio-E5 (Seeed)

Day 2 · LoRaWAN to TTN



- **STM32WLE5JC**: Cortex-M4 + SX126x LoRa radio
- LoRaWAN, EU868 / US915
- UART with AT commands (set to 19200 baud for the Nicla)

TALK

Arduino UNO Q

Day 4 · an LLM running on the board



- **Qualcomm QRB2210** quad-core, Debian Linux
- **+ STM32U585** microcontroller for real-time I/O
- 2 or 4 GB RAM · Wi-Fi 5 · Bluetooth 5.1

Made at the ICTP FabLab

Custom PCB: connects the Nicla Vision to the Wio-E5 LoRa module over UART

3D-printed case: houses the Nicla, the Wio-E5 and the PCB as one node

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